



Safecracking

Manipulating Mechanical
Combination Locks

by Ammon Franklin

SAFE cracking: Manipulating Mechanical Combination Safes

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INTRODUCTION

There are many ways to get into a mechanical lock safe. Peeling a safe is the process of tearing it apart. Explosives, acetylene torches, or thermic lances can penetrate the safe. Drilling is meant to cause the least external damage while opening the safe. Drilling may allow access to a punch in order to disable the lock, or it may allow the use of a scope to “spy” the combination without disabling the lock. Radiological penetration, one of few nondestructive safecracking methods, can allow a safe-cracker to see lock operations through the safe walls.

In this e-book I describe the non-destructive process of manipulation: using touch to determine the combination of a mechanical lock. Successful manipulation of a dial lock determines the lock combination without any damage to the safe.

Before anything else, watch the [safecracking video of Jeff Sitar](#) (“Jeff Sitar - Champion Safe Cracker”).

Watched it? Good! You're not him. You'll never crack a bank vault in 5 minutes. Your first safe, if it's in ideal condition, will take you dozens of hours to crack. Safe-cracking requires tenacity. If you don't want to stick with it, then you're better off quitting now.

Still here? Good!

The purpose of this e-book is to teach you about dial combination locks, their weaknesses, and how to exploit them. YOU are solely responsible for how you use this information. I have made every attempt to concisely describe the process of manipulation you'll need to learn to open a locked dial combination safe. Yet, you may want to read this e-book several times before beginning your practice.

PART I: UNDERSTANDING DIAL LOCKS

In Part I we explore the parts and function of dial locks. Until you understand

the vocabulary and mechanics of safe locks, you will find safecracking all but impossible. Figure 1 represents the internal parts of a typical UL2 dial combination lock.

Apart from the lock itself, most safes include a boltwork of some sort. The dial lock does not unlock the door, it unlocks the boltwork, which are retracted using the exterior lever or wheel.

Vocabulary and Function

Bolt (lock bolt): The portion of the mechanical lock that stops the safe door handle or wheel from operating the main locking bolts on the safe door.

Boltwork (main locking bolt[s]): The mechanical portion of the door, operated by the handle or wheel, that physically locks the door.

Cam (drive cam, wheel cam): The portion of the lock that is fixed to the spindle and dial, which has a gate for the nose to hook into.

Cam Gate: The portion of the cam that receives and catches the nose when the fences line up over the wheel gates.

Change Index: The mark beside the dial that indicates which number is being dialed *during the combination change process*. Usually the change index mark is at roughly 300° , or at 85 when the dial is zeroed on the open index. The change index is usually a single mark.

Contact Points: The left and right contact points are where the nose comes into contact with the respective edges of the cam gate. Between these points the nose does not touch the cam, but the fence does touch the wheels. See Figure 2, below.

Dial: The face of the lock, showing 100 graduated positions (usually marked 1-100 or 0-99). Less commonly there may be more or less numbers on the dial, e.g., 80 or 130.

Dial Ring: A plate surrounding and behind the lock dial. The dial ring typically has change index and open index marks.

Drilling: The process of putting a hole in a safe in order to punch the lock or spy the combination.

Drive Pin: A small protrusion on the face of each wheel that contacts the wheel fly of the adjacent wheel, causing it to turn in time.

Fence: a small bar resting across the tops of the wheels in the wheel pack. The fence drops when the gates of each wheel are aligned under it.

Fence Depth: The drop of the fence when the nose is in the cam gate (between the contact points). The lower the fence depth, the smaller the distance between the contact points. The greater the fence depth, the greater the detail may be determined about the wheel pack.

Gate: The gap in each wheel that the fence fits into when they are all aligned. Also, the gap in the cam that the nose fits into.

Gate Position: The position of the cam when the nose is between the contact

points and the fence is resting on or in the wheels.

Lever: The portion of the lock that connects the lock bolt with the fence and nose. Also, the door handle or wheel.

Manipulation: The systematic process of determining the combination to a mechanical dial lock using one's physical senses.

Nose (lever nose, lever hook): The nose is the hook attached to the lock lever that grabs the cam gate when the wheel gates are aligned under the fence. The nose allows the dial to pull the lock bolt away from and release the boltwork.

Open Index: The mark above the dial that indicates which number is being dialed. Nearly always the open index mark is at 0°, or at 85 when the dial is zeroed on the open index. The change index is usually a single mark.

Parking: The placement of a wheel in a fixed position while other wheels are manipulated.

Peeling: The physical process of tearing a safe apart.

Punching: Punching a safe involves drilling a safe and using a punch or rod to disable the lock bolt.

Radiological Penetration: Using x-ray technology to see the lock through any external metal plating of the safe.

Relocker: A device that locks the safe door when the safe is under attack. Relockers may be included in the mechanical lock or be supplemental to the bolt-work.

Slop: The difference in size between the fence and the wheel gates, generally allowing from $\pm.75\%$ to $\pm 1.25\%$ dialing error tolerance. For example, if the combination is 77-33-55, dialing 76-33-55 will still open the safe.

Spindle: The rod connecting the dial on the outside of the safe with the cam inside the safe.

UL1: A lock classification indicating resistance to manipulation for 20 hours. A Group 1 lock as classified by the Underwriters' Laboratories.

UL1R: A lock classification indicating resistance to professional manipulation for 20 hours and shielding from radiological penetration.

UL2: A lock classification indicating resistance to professional manipulation for 2 hours. A Group 2 lock as classified by the Underwriters' Laboratories.

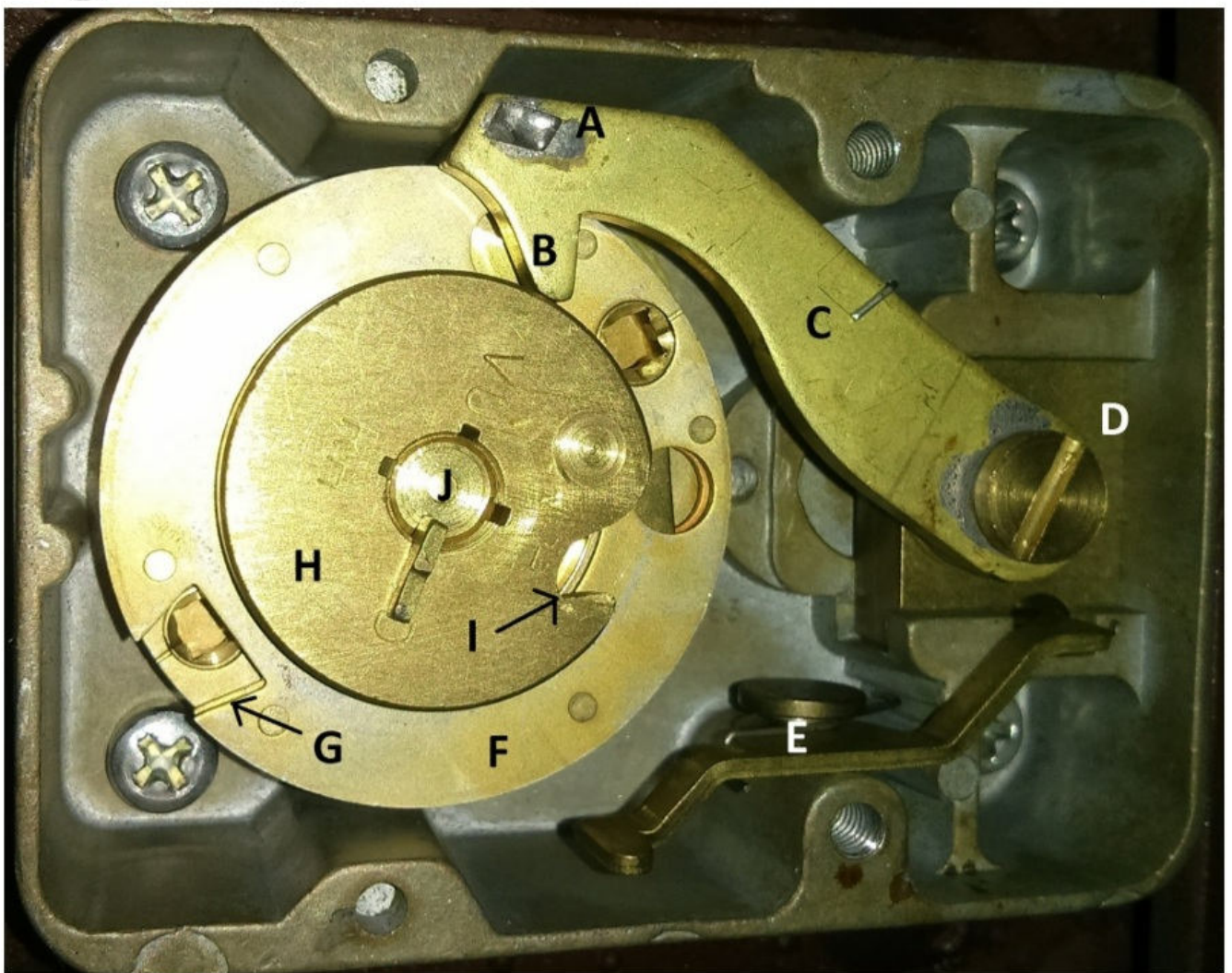
Wheels*: The independently rotating disks of the wheel pack. Each has a notch (gate) to receive the fence. The first and second wheels have both a wheel fly and a drive pin. The third wheel has only a wheel fly. (In four-wheel locks, the first, second, and third wheels are mechanically identical, while the fourth wheel has only a wheel fly.)

Wheel Fly: The portion of the wheel that contacts and is pushed by a drive pin, causing the wheel to turn.

Wheel Pack: The collection of wheels in the dial lock, each representing a digit of the lock's combination.

*Some people call the wheels in the wheel pack “tumblers”. This is just stupid. Don't do it.

Figure 1



A. Fence (extending behind the nose)

B. Nose

C. Nose Lever

D. Lock Bolt

E. Relocker (“E” placed over pivot point. When the left half is depressed, the right half releases the lock bolt

F. Wheel One

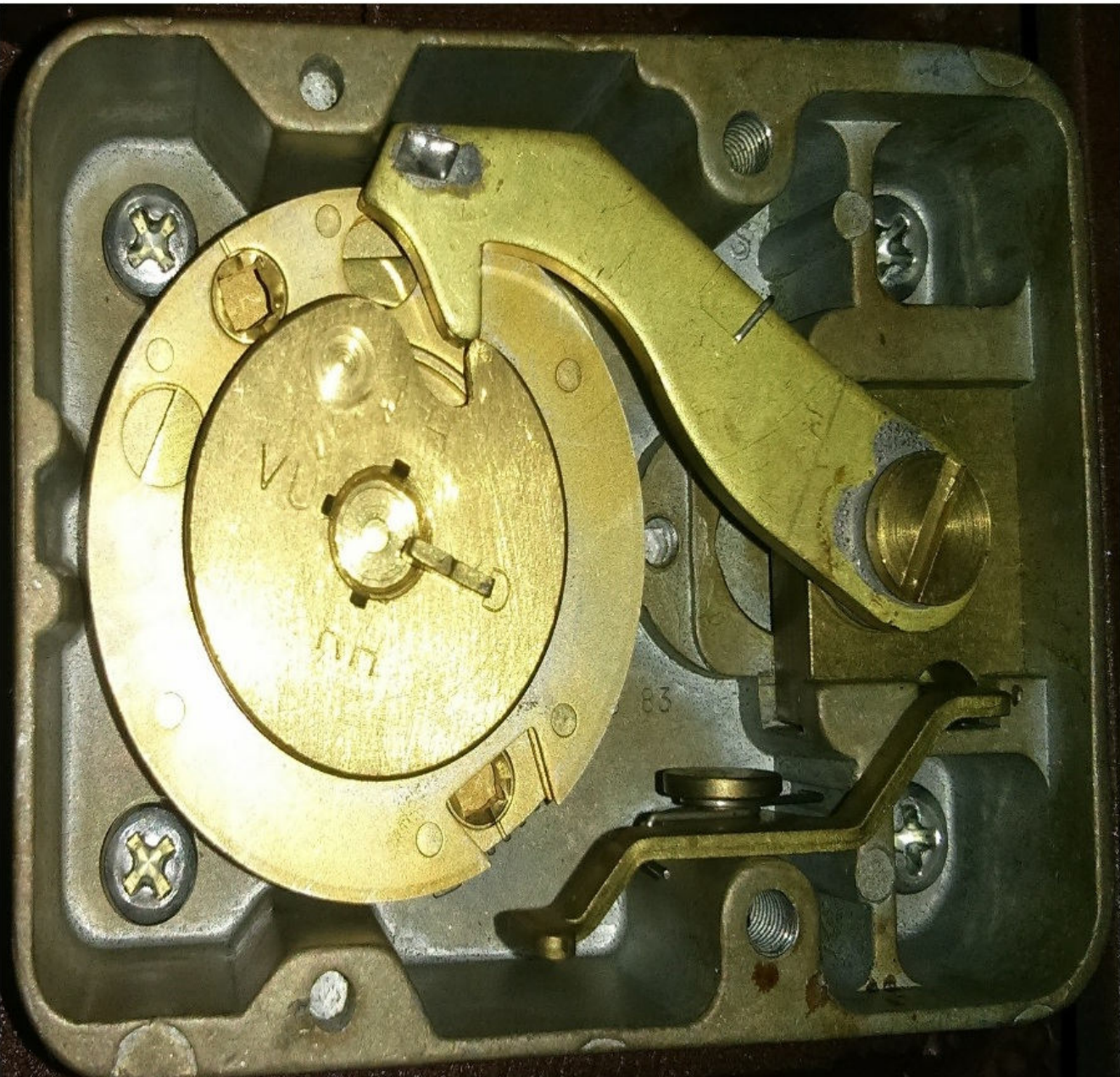
- G.** Wheel Gate on Wheel One
- H.** Cam Wheel
- I.** Cam Gate on Cam/Drive Wheel
- J.** Spindle (with keyway visible between H-I-J)

FIGURE 2. The dial is turning clockwise. This view from back of lock shows cam moving counterclockwise/anticlockwise.

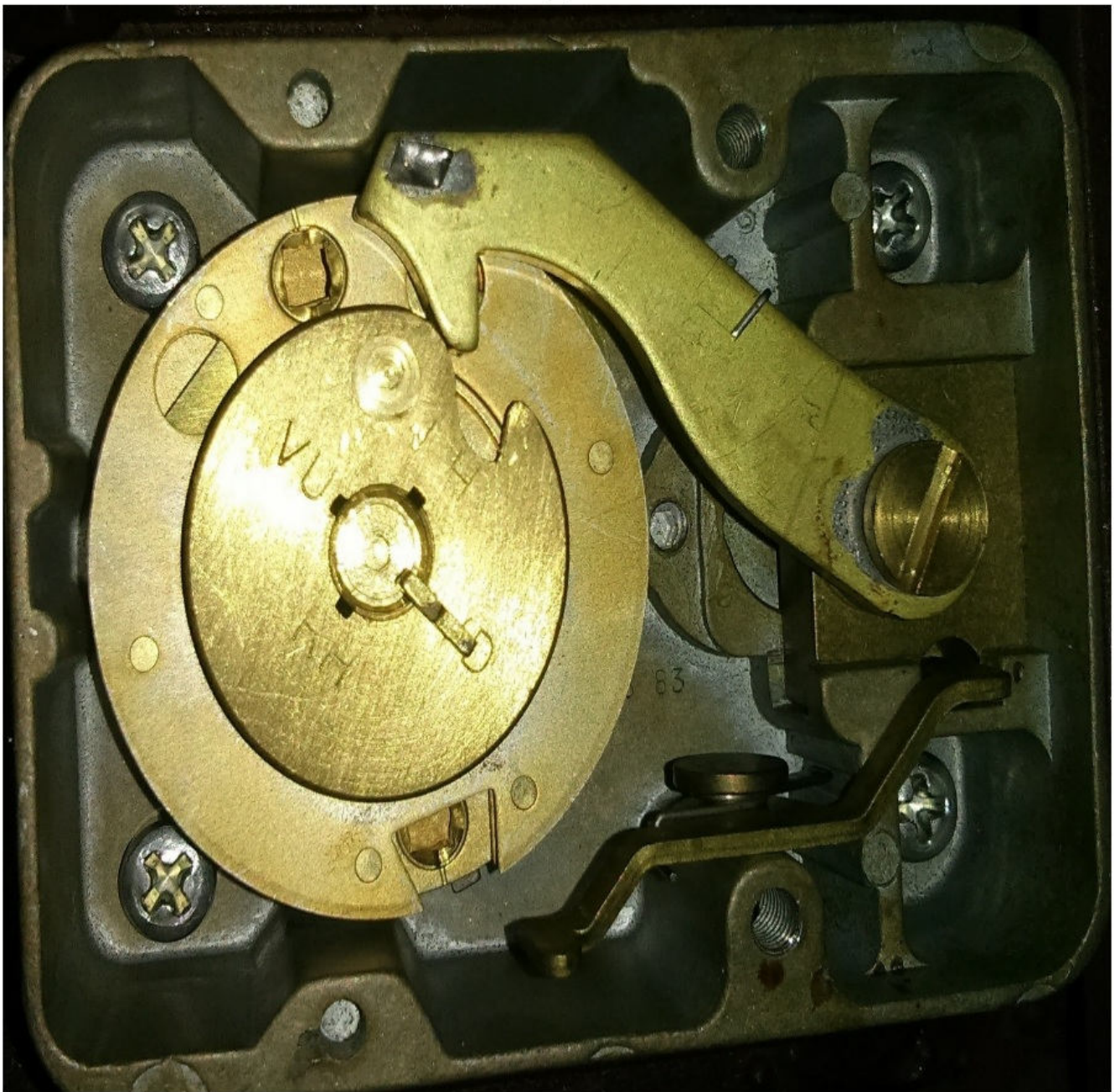
B: Left contact point. The nose drops slightly and the

fence rests on the wheels. This the moment just

A. The cam gate approaches the nose. before the nose leaves contact with the cam.



C. right contact point. The cam moved under the suspended nose until making contact once again with **D.** the nose is raised by the edge of the cam gate, the cam gate. lifting the fence off the wheel pack.





Abbreviations

For the purpose of simplifying this text and avoiding confusion, I use a number of abbreviations. I also assume a three digit combination.

C1: The first digit of the combination, as controlled by the first wheel of the wheel pack.

C2: The second digit of the combination, as controlled by the second wheel of the wheel pack.

C3: The third digit of the combination, as controlled by the third wheel of the wheel pack.

R: Dialing right / turning the dial clockwise (dial numbers descending relative to the open index).

L: Dialing left / turning the dial counterclockwise/anticlockwise (dial numbers ascending relative to the open index).

W1: The first wheel of the wheel pack, which represents the first digit of the combination.

W2: The second wheel of the wheel pack, which also represents the second digit of the combination.

W3: The third wheel of the wheel pack, which represents the third digit of the combination

Classification of Dial Locks

Dial locks of the sort used in safes are grouped in two categories, Group 1 and Group 2. Since the late 19th century, Underwriters' Laboratories has oper-

ated as a safety science company rating safes and safe locks. Basic UL dial lock ratings include UL2, UL1, and UL1R (see “Vocabulary” for definitions).

Beginner safe crackers would do well to practice on standard UL2 locks, such as the Sargent and Greenleaf 6730.

In order to make manipulation more difficult, UL1 locks include features not discussed in this e-book, such as false gates, rounder wheels, and devices that shield the safe cracker from feeling the contacts points.

Dialing Order and Operation

Our example combination for this e-book is 77-33-55, with a gate position (wheel cam gate) within 6-13.

Again, dialing right (R) means turning the dial clockwise, and dialing left (L) means turning the dial counterclockwise/anticlockwise.

The most common dialing order is L-R-L. The dial is turned four times L, stopping at the first combination digit of

77. The dial is then turned R , stopping the *third time* you see 33 at the open index (this may not be three full turns). Turn the dial L again, stopping the second time you see 55 at the open index (this may not be two full turns). Turn the dial R until the fence drops into the gates (at the gate position). To fully retract the lock bolt, keep turning the dial until it comes to a hard stop (roughly 5-10 digits past the gate position. The safe is now unlocked..

Some locks may have R-L-R dialing orders. Or in the case of four digit combinations, R-L-R-L or L-R-L-R dialing orders. Please note that the stopping position of the dial is not considered a digit of the combination. Also see Tip #8.

The interior operation of the lock is as follows (assume the dial has been turned four full turns L) :

- A. As the dial spins, so do the spindle and cam. They are in fixed position relative to each other.
- B. First rotation R -- The cam turns with the dial to the right. No wheels turn.

- C. Second rotation -- The second rotation includes the W3 with the cam. As the cam completes its first full rotation R, the cam's drive pin contacts the wheel fly on W3 of the wheel pack and causes it to begin turning along with the cam.
- D. Third rotation -- After the second full rotation, the drive pin on W3 will contact the wheel fly on W2, which includes W2 in the third rotation.
- E. Fourth Rotation -- After W2 completes its full rotation, the drive pin on W2 contacts the wheel fly on the W1, bringing it into rotation with the other wheels. At this point, the entire wheel pack is rotating together and will continue to do so as long as the dial is turned to the right.

The interior operation of the lock as you dial the combination is as follows:

- A. From the fourth rotation L, as the dial continues to turn L, all

of the wheels in the wheel pack are rotating together. When you stop on the first combination digit, 77, all of the wheels are positioned at 77. Since only the W1's gate is at 77 the fence won't drop.

- B. Turning the dial R first turns the cam, then adds W3 and W2 with subsequent rotations. Stop the dial the third time you see C2 (33) align with the open index. You have not yet turned the dial far enough to move W1 from its position at 77. The gates of W1 and W2 are now aligned under the fence.
- C. Turning the dial L moves only the cam for the first 360° . You've seen 55 pass the open index one time. The cam then picks up W3 and rotates until you stop at C3 (55). Once again, you have not continued turning the cam and W3 past 720° , so W2 and W1 have not be moved from position.
- D. All three gates are in position under the fence, but the lock

will not open until the nose is also over the cam gate (gate position). See Figure 3, below. Turning the dial/cam R from its last position at 55 leaves the entire wheel pack in position. The cam continues R until the cam gate aligns under the nose (roughly at 10, in our case), when the fence drops into the gate and the nose hooks on the cam. There is a distinctive feel as this happens. Continuing R approximately another 5-10 digits past the gate position retracts the lock bolt, releasing the main locking bolts. The lock will come to a hard stop as the lock bolt retracts fully. It will turn no further to the right since the lock bolt is stopped, the nose is locked into the cam gate, and the fence is dropped into the gates of the wheel pack.

Turning the dial back to the right at least four full turns does the following: first, draws the nose out of the cam gate: second, moves W3 from its posi-

tion at 55; next, moves W2 from its position at 33, and finally moves W1 from its position at 77. The wheel pack has been cleared.

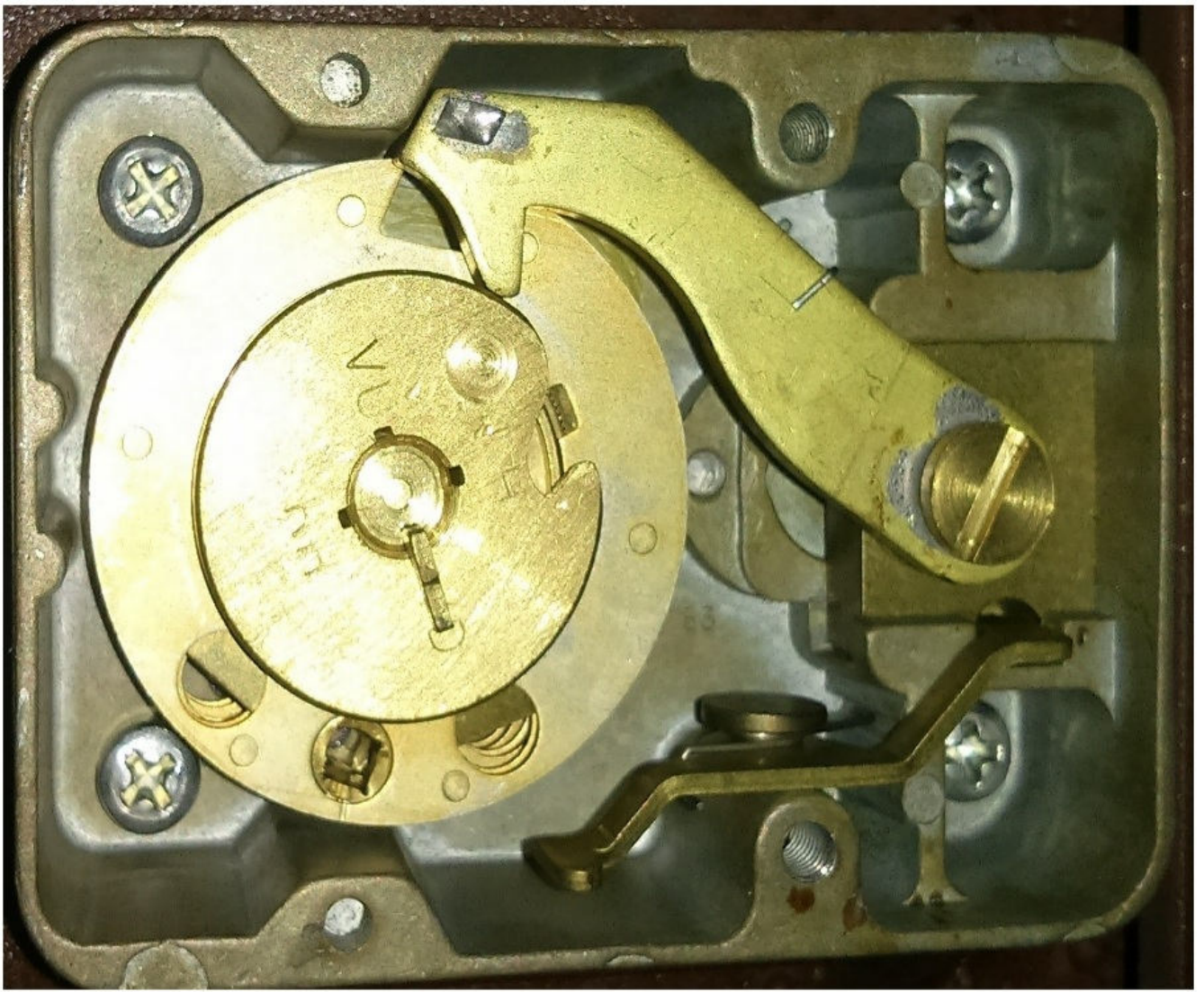
Failing to turn the dial four *full* turns to right after closing the safe will leave one or more of the wheels with their gates under the fence. A safe cracker can feel the positions of the wheel and easily devise the remainder of the combination. Always, *always* turn the dial four full turns after closing your safe.

Understanding the operation of the lock is key to devising the combination *and* to saving time or effort while doing so.

FIGURE 3. The wheel gates are aligned under the fence. This series of photos shows the subsequent aligning of the cam gate under the nose. Images A-G.

A. Showing the wheel pack's gates aligned under the fence (hidden behind the upper part of the nose). **B.** The cam is rotating counterclockwise/anticlockwise, with the cam gate approaching the nose.





C. The nose begins to enter the cam gate as the fence enters the wheel gates.

D. The nose continues to enter the cam gate.



E. The nose has fully entered the cam gate, as also

F. The cam begins pulling the nose lever, which the fence has entered the wheel gates.

subsequently begins pulling the lock bolt left.





G. The cam has pulled the nose lever and lock bolt to the fully retracted position.



PART II: DIAL LOCK WEAKNESSES AND FAILURE

It is the goal of the safe cracker to exploit the weaknesses of the lock or safe. Basic UL2 locks have a number of features that allow a safe cracker to feel what happening inside the safe, or to shorten the time necessary to manipulate the safe.

Slop

Slop is the difference between the width of the fence and the width of the wheel gate. With a UL2 lock there is typically a $\pm 1.25\%$ error tolerance. This means that if you intended to dial your combination of 77-33-55, but instead you dial 76-32-55, the lock has a good possibility of still opening. With a gate tolerance of 1.25% on each side of the fence, the range of successfully dialed numbers consists of a 2.5 digit gap. The safe manipulator no longer has to try combinations at single, whole number intervals. Rather s/he need only try numbers at intervals 2.5, 5, 7.5, 10, 12.5, etc. A three digit combination lock with a 100 digit dial has 1,000,000 possible combination; however, due to

slop, there are effectively only 64,000 possible combinations.

Some lock models, such as UL1 locks, may have a $\pm .75$ error tolerance.

Gate Position

As the cam rotates the the nose of the lever rests (or is held by a spring) against the outer surface of the cam and the fence is suspended slightly above the wheel pack. When the gate position rotates under the nose, the nose drops slightly allowing the fence to rest on the wheel pack for that short period of the dial rotation. If our gate position is centered at ~ 10 , the fence may rest on the wheel pack between 6 and 13, for example. As the dial is turned to the right you can feel the nose drop at 13, and “bounce” back up 7. ,The left “contact point” is at 7, and the right contact point is at 13. In this gap you can feel slightly different features of the wheel pack, depending on the myriad possible relationships each wheel may have with the others. You can feel these features because the fence is resting on the wheel pack.

Wheel Irregularity

Each wheel in the wheel pack may have slightly different features. An ever-so-slight ovalness of a wheel may be perceptible to the safe cracker. These irregularities allow the safe cracker to better understand what's happening in the lock as it's manipulated.

Third Digit Position

Because the cam has a gate position around 10, any C3 should not be in the gate position. The closer C3 is to 10, the more likely the occurrence of interference. For example, if my C3 is 11 and my gate position is near 10, I will be turning L to my C3 at 11, then I'll turn R only one digit before the fence drops. If it doesn't drop, then I'll continue another 360°, but I'll move the 11 before I get to the 10. With C3 (11) now moved to 10 the fence again may not drop. Because of this possible interference, factory combination would not a C3 within 10 digits of the gate position. In our example, this would preclude a C3 gate at numbers 0-20.

This further reduces the number of possible combinations 20% on C3, or from 64,000 to roughly 51,200.

HLH and LHL Combination Patterns

Factory combination formations have other restrictions. In addition to the restriction that the third digit not be proximal to the gap (mechanical reasoning), strictly ascending or descending combinations are not considered good practice. Thus, all of the following combinations would be proscribed:

33-55-77	77-55-33
77-55-10	55-75-20
25-50-75	30-20-05
95-10-15	

Again we have a reduction in effectively distinct combinations from 51,200 to only 22,330. This is merely 2.3% of the original 1,000,000 combinations for a three wheel lock with a 100 digit dial.

PART III: YOUR LOCK

Assessing the Condition of the Lock

Locks are mechanically simple, although lack of maintenance and wear over time can cause them to malfunction. The majority of problems with safe locks are caused by friction.

Individual wheels are designed to rotate around the spindle without interference from the spindle itself. Wheels also rotate independently of other wheels. An incorrect or degraded lubricant can cause wheels to seize or stick periodically. In addition, wheels may be worn through long use or jostle while the safe is in transport. If a wheel begins to lean--causing contact with adjacent wheel(s) when their respective drive pins and wheel flanges are not in contact--this contact may cause inadvertent movement of wheel position.

Dirty or malfunctioning dials, spindles, and/or wheels may also hamper the safe cracker's ability to feel what's going on in the lock.

Record Keeping and Accuracy

Good records, careful dialing, and consistent lighting are key to successful

manipulation of safe locks. Sloppy dial turns can waste time at the best, and cause you to miss the combination entirely at the worst.

Plan to dial with great consistency under uniform lighting. When you dial a digit, consider turns in terms of *tenths of a digit*. If you intend to dial 75, but dial 74.6 instead, you may have a different result and end with inaccurate information.

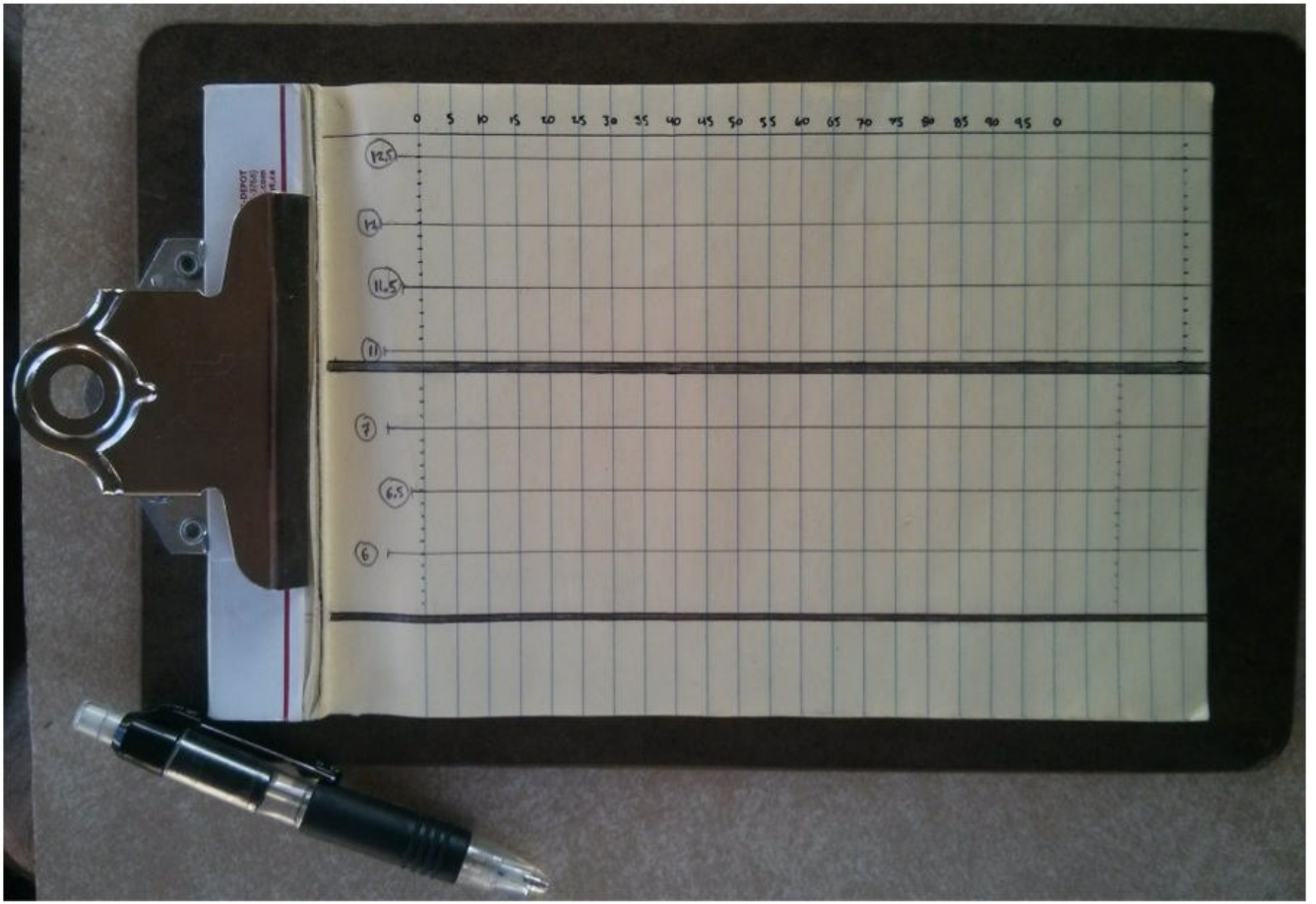
Turn the dial as smoothly as you can. Smooth dialing allows you to feel the features of the cam and wheels better.

Ensure that your lighting is consistent. Changes in natural light at different times of day can change how you perceive the tenths of each digit on the dial (see tip #6, below). This slight change in perception due to changing light can affect your knowledge of changes in contact points.

During the systematic process of determining your safe's combination, you will record contact point data (edges of the gate position) for each number or combination that you dial. This is most

commonly recorded in graph form. I suggest using a sheet of graph paper in landscape orientation. The horizontal length of the paper (X axis) will represent the combinations dialed (0, 2.5, 5, 7.5, 10, 12.5, 15, etc.), and the vertical or Y axis will represent the gate area in tenths of a digit. Using the lines of the graph paper as fourths of a digit, and marking points on or in between the lines may work for you. See Figure 4, below.

FIGURE 4. The author's preferred record, a half-sheet-sized note pad and clipboard. X-axis in five-digit increments (data also recorded between these points). The upper half of the sheet is the right contact point (11-12.7 in 1/10th digit increments), and the lower half the left contact point (5.5-7.5 in 1/10th digit increments).



Mind the Gap

Your first job is to determine the gate position. This is often around 6-8 digits wide and located in the dial range of about 85 to 15. Standard UL2 Sargent & Greenleaf locks often have the gate position at about 6-12 (left and right contact points, respectively).

A good way to find the gate position is to turn all wheels to 50 (four or five turns left, for our example). Then slowly turn the dial to the right. This allows you to turn the cam wheel without interference of the wheel pack. As you pass through the expected cam gate range, try to feel the contact points “drop” and “pop.” On some locks locat-

ing this gate range is an obvious and easy task. On other locks, this may be a difficult nonetheless essential task.

Note the location of the gate position. You'll be returning here often.

Determine the Number of Combination Digits

Your next job is to determine the number of digits in your lock's combination.

You may begin either direction, but for our example we will turn the dial *five full turns* L (five because of the possibility of a four-digit combination). Stop with the number 50 at the open index. Next, slowly turn R. Recall that you have left all of the wheels in the wheel pack at 50. With the completion of each full turn, as you pass the number 50, you should feel the cam wheel "pick up" each additional wheel. Picking up wheels causes a slightly "heavier" feel in the dial. You may possibly hear a click as each drive pin contacts the wheel fly of each successive wheel, although I find this is usually not the

case. Your refined sense of touch should tell you all you need to know.

Each full turn, with each additional “heaviness” added to the wheel, represents one digit of the combination. A three digit combination will feel additional resistance in the dial with the end of the first, second, and third full dial turns. A fourth, fifth, etc. will have no effect on how the dial feels. A four digit combination will feel additional resistance at the end of the fourth full dial turn, but not the fifth, sixth, etc.

Lock Problems

In your attempt to assess the qualities of your safe lock, you may find certain inhibiting factors: heavy friction, internal malfunction, or bent spindle. Some of these problems are impossible to remediate.

It’s nearly impossible to clean the wheel pack when the safe is closed and locked; however, if dirt or grime is behind the dial or on the outer parts of the spindle, you may be able to clean it. One way to clean the dial and spindle is to use a

small, flexible syringe nipples with solvents to free up friction on the dial or spindle. Different lock and dial designs may inhibit cleaning.

If wheels in the wheel pack are stuck, or contacting each other, you may not be able to determine the number of wheels in the wheel pack, thereby finding the number of digits in the combination. Spinning the dial hard, warming the dial/lock up, or just give the lock some use (continued left/right rotations), may free up the wheels. Spinning the dial too hard, however, may cause additional problems.

A bent spindle may incapacitate your manipulation efforts entirely by causing the wheel pack to wobble as the dial turns. This may occur whether or not you're turning any of the wheels in the wheel pack. If the spindle is bent, you may feel consistent friction on one side of the dial with every turn. You may also see the dial wobble as you turn it.

NEVER, EVER mess with the dial by hitting it, banging it, or prying on it. Doing so may turn your safe to scrap metal.

PART IV: GETTING DIGITS

By this point you should already have located the gate position and determined the number wheels.

Slipped Wheels and Guessing the Combination

If you have no idea what the combination to the safe could possibly be, then you can skip this section. If you knew the combination, but it no longer works; or you know the person who originated the combination, then read on.

Your combination no longer works. On locks that can have the combination changed, a wheel may have slipped from one position to another adjacent position due to an overly rigorous turn of the dial. For example, if your combination was 77-33-55, and the 55 may have slipped to 53. Thus dialing a 55 would not open the lock. The final digit of the combination is the most likely to have slipped, so begin by trying

proximal combinations: 77-33-53 and 77-33-57. You may try adjacent combination numbers on wheels three and two, but change only one digit at a time.

If you knew the person who originated the combination, you may try numbers associated with this person, such as birthdays, anniversaries, and important family dates. People often use digits familiar to them or that are easy to remember. An educated guess can go a long way. About 20% of safes cracked by the author have such combinations (e.g., 33-66-33, 75-25-50, 22-11-63, etc.)

Method for Determining a First of Three Digits

Remember the slop. That's going to save you a lot of time! So will accurate dialing and record keeping.

For this tutorial, we're going to assume the gate position is roughly at 6 to 12, as is common with many S&G UL2 locks. Begin by turning L four full turns. Continue to and stop at 0. All three wheels are at 0. Turn R to the gate

position at 10. This is a turn of only 90 digits, NOT a full turn. Move the dial back and forth carefully in the gate position. Measure contact points in the gate position (e.g., 6.5 and 12.1). Record this on your tracking sheet (See Figure 4).

Turn L from the gate position to 2.5, which will move all three wheels from 0 to 2.5. Again turn R to the gate position at 10. Measure the gap in the left and right contact points. Record the contact points.

Turn L to 5, which will again move all three wheels from 2.5 to 5. Turn R to the gate position and measure the gap in the contact points. Record the data.

Testing 7.5 may get trickier. When checking digits in the gate position, you may have to get creative. In this case you may turn R to ~6 and record only the left contact point, then *continue* R nearly a full turn until you get to 10, where you can then record the right contact point at ~12. Just be sure not to pass a 360° turn R from 7.5, or you'll move W3 from its position at 7.5.

You can repeat this tricky process when testing 10, and possibly 12.5. Just be sure to record as much data and as accurately as you can.

Continue at intervals of 2.5 by testing 12.5, 15, 17.5, ... 92.5, 95, and 97.5.

As you review your data, you're looking for a point or points in the series of tests where the gap between the left and right contact points becomes smaller. This happens when the wheel gate of the largest (or most raised) wheel is under the fence. This allows the fence to rest slightly lower than adjacent numbers while the nose is "hanging" in the cam wheel gate. When the fence is lower, the nose contacts the left and right portions of the cam wheel gate slightly sooner, making the measured gap smaller.

You may also be lucky enough to note two or more points in your data that may indicate wheel gate positions.

This does not, however, tell you the precise combination digit, nor which wheel is causing this change.

To narrow down from a range to a single combination digit, repeat the above test with every digit in the prospective combination digit area. In our example, we have a narrow measurement at 77.5 and 80. You should test every digit around 73-83 or so. Again, the narrowest gap measurement is likely where the combination digit is located.

Determining on Which Wheel Your FIRST Prospective Combination Digit Is Located

Ok, now you have a prospective combination digit of 77. You don't know which wheel this combination digit represents, so you'll dial the following combinations and measure the gate position after each one. We get the number 67 by taking 77 minus 10. We get the 87 by taking 77 plus 10. A ten digit difference should allow sufficient variation in measurement to determine further information.

Figure 5. Testing to determine which wheel has a potential gate situated under the fence.

Combination (W1-W2-W3)	Left Contact Point	Right Contact Point	Contact Point Spacing
77-77-67	6.3	11.9	5.6
77-67-77	6.3	11.9	5.6
67-67-77	6.1	12.0	5.9
77-77-87	6.3	11.9	5.6
77-87-77	6.3	11.9	5.6
87-77-77	6.1	12.0	5.9



Remember that in our first test we were placing ALL of the wheels in the wheel pack at a given number. And our test determined that we had a possible combination digit when ALL of the wheels were at position 77. By leaving all but one (two out of three) of the wheels in the position at 77, and changing a third, we can see if there is a measurement change from what we found with our first test measurement with all wheels at 77. By testing a variation both above and below 77 (67 and 87) we can double check our data.

As we see in Figure 5, when W1 is changed to 67 or 87, there is a notable increase in the measurement between the left and right contact points. By this we know that the gate of W1 is at 77.

Determining a Second When You Have the First Combination Digit

Continuing our example from the previous section, we assume we have the first combination digit of 77. By carefully leaving the third wheel (first combination digit) in place, we can change the position of the other wheels and measure the contact points. This can potentially save the manipulator hundreds of turns of the dial.

Leave the first digit in at 77, and move the other wheels to 0, 2.5, 5, 7.5, etc. Carefully record your measurements. As you find an additional wheel gate position on wheel one or two, follow the previous process of narrowing down the possible gate position to a specific digit. In our case, this is 34.

Now, repeat the test to determine which wheel has a gate at 34 by dialing the following:

77-34-44

77-44-34

77-34-24

77-24-34

Again, when the second or third digit is moved from its position at 34, the measurement between the contact points will widen. You know then that the moved wheel is where the 34 belongs. In our example, this is on wheel one (the third combination digit).

Determining a Second When You Have the Second Combination Digit

If you have the second digit of a three digit combination you will need to dial all three wheels in a similar pattern to the previous section. For our example, let's assume that we know the second digit is 34, but do not know the first or the last digits.

Begin your test by dialing 00-34-00, followed by 02.5-34-02.5, 05-34-05, etc. Each time being sure to dial care-

fully and record your data equally carefully.

Once you have a prospective digit, repeat the test previously described pattern to find which wheel your prospective combination digit is on.

Determining a Second When You Have the Third Combination Digit

For this example, let's assume you only know the final combination digit, and it is 55. As in the previous section, you will need to use three digit test combinations as follows: 00-00-55, 02.5-02.5-55, 05-05-55, etc.

Again, use the test to determine which wheel a prospective combination digit may be on. In our example let's assume we've found a second digit at 34, you would use the following test combinations:

24-34-55

34-24-55

44-34-55

34-44-55

Determining a Third Digit of a Four-digit Combination

You would use the same pattern as previously described for finding additional digits, even if the combination is 4 or even 5 digits long.

Determining the Last Digit

Use a brute force attack. You know two of the three combination digits, and which wheels they are on, so you only have 40 possible combinations to test. I generally like to test every two digits, rather than every 2.5, just to increase my error tolerance. For example, I may be off by one digit on the combination digits I assume I have. By dialing the unknown digit off by 0.25% may cause the lock to not open. And by moving from a 2.5 digit interval to 2 digit interval only increases the number of combinations to test by 10, from 40 to 50.

PART V: TIPS AND TRICKS

Tip #1: The better you understand how locks function, the more time and effort you can save. Always be aware of which wheels are where, and which are moving at any given time.

Tip #2: If you succeeded in opening your safe or changing your combination, *always* test your combination several times *with the door open!* You can easily lock the door open by turning the dial left, and extending the door bolts with the door handle. Be careful not to try closing the door with the bolts extended! You may inadvertently damage your safe. When testing your combination, try adjusting each combination digit one at a time. Finding where your combination no longer works will give you the center of the wheel gate. For example, you think your combination is 77-34-55:

78-34-55 - works

76-34-55 - works

75-34-55 - doesn't work

79-34-55 - doesn't work

You have confirmed the first digit is 77.

77-35-55 - doesn't work

77-33-55 - works

77-32-55 - works

77-31-55 - doesn't work

You have discovered your second digit is actually 33, not 34 as you first thought.

77-33-56 - works

77-33-57 - doesn't work

77-33-54 - works

77-33-53 - doesn't work

You have confirmed the final digit is 55.

Tip #3: You may *think* you have a combination digit correct, but only be at a point in the wheel pack where the fence depth is greater (or in a false gate). That's fine; the greater the fence depth, the more information you can get. Park the wheel at its greatest depth

and retest the other wheels. You may progress in stages as you increase the fence depth.

Tip #3: If you're lucky enough to get C1 and C2 before C3, testing for the last should only take a matter of a few minutes.

You don't need to redial full combinations each time as you brute force test for the final digit. Just leave W1 and W2 in place (park them) while turning back and forth between your various C3 test digits (on W3) and the gate. When you hit the right digit (or close), the fence will drop.

Tip #4: As with tip #3, if you know only the first digit, you can test the last two digits by parking W1. This takes a bit more practice and careful attention to what's happening inside the lock.

Tip #5: Depending on your safe and lock designs, putting pressure on the door handle or wheel may transfer pressure through the lock bolt and press the nose to the wheel cam (or fence to the wheel pack). As long as you do it consistently, this may en-

hance your ability to measure changes in the gap between the contact points. If you do it inconsistently, some your measurements will certainly be off, and you'll have to start over.

Tip #6: You may come up with your own system depending on the dial you're working with, but I don't actually divide the the space between each digit mark on the dial into equal tenths. I use approximations based on the visual relationship between the dial digit mark and the open index mark. For example, if my left contact point is somewhere between dial numbers 6 and 7 and my right contact point is between 11.4 and 12.6:

6.0 - The digit mark and open index mark are in direct vertical alignment

6.1 - The digit mark is slightly misaligned to the right of the open index mark

6.25 - The left edge of the digit mark is aligned with the center of the open index mark

6.3 - The digit mark is still overlapping the open index mark, but just barely

6.5 - The open index mark is perfectly aligned between digits 6 and 7.

6.7 - The open index mark is just barely overlapping onto the left side of the 7.

6.75 - The left side of the 7 is aligned with the center of the open index mark

6.9 - The 7 is nearly fully aligned with the open index mark.

11.3 - The digit mark is still overlapping the open index mark, but just barely

11.4 - The open index mark is just barely off center.

11.5 - The open index mark is perfectly aligned between digits 6 and 7.

11.6 - The open index mark is just barely off center

11.7 - The open index mark is just barely overlapping onto the left side of the 7.

11.75 - The left side of the 7 is aligned with the center of the open index mark

11.9 - The 7 is nearly fully aligned with the open index mark.

12.0 - The digit mark and open index mark are in direct vertical alignment

12.1 - The digit mark is slightly mis-

aligned to the right of the open index mark

12.25 - The left edge of the digit mark is aligned with the center of the open index mark

12.3 - The digit mark is still overlapping the open index mark, but just barely

This kind of system allows you to easily perceive discrete differences in contact point measurements without being constrained by precise 1/10th sections of dial digits.

Tip #7: Using a stethoscope *may* help; however, if your lock makes noise, it's probably enough contact to feel through the dial to start with. You can get a cheap mechanic's stethoscope at many auto parts stores.

If you want to read a bunch of crap about using a stethoscope to crack a safe, read the [WikiHow on safecracking](#). I suspect it was put there by a trained safe technician to screw with people by playing on their Hollywood perceptions of safecracking.

Tip #8: When you understand the operation of the wheel pack, you'll realize that the combination may be manipulated both L-R-L and R-L-R. The fence will drop when the nose enters the gate position, but the lock bolt can only be retracted by dialing a single direction--usually R. Dialing a combination backwards will require a switch in dialing direction once the gate position is reached. Depending on the lock design, the L-R-L and R-L-R combinations may be slightly different.

Tip #9: When the fence is over one of the gates, and the nose drops slightly lower in the cam gate, the right contact point will show a more marked difference than will the left contact point. Unfortunately, the right contact point is also more difficult to measure accurately. This phenomenon is due to the sloped nature of the right contact point. Thus, you should be the most careful in measuring the right contact point.

PART VI: FAILURE!

You may consider quitting after many hours without success. If, however, you have made progress in fence depth, a shortening in the distance between contact points, then you will eventually succeed. See Tip #3. Park the wheel or wheels you know are “low” and retest the other(s). Be tenacious, be persistent!

After great effort, if you still have had zero success in determining the contact points, or in increasing fence depth, then you may have a more complicated lock. Some locks, including some UL2 locks use mechanisms to hide the details of the wheel pack. At this point, you may have a lock that cannot be manipulated, let alone by a beginner.

PART VII: SUCCESS!

Congratulations on cracking your safe!
That feeling when the fence drops into
the gates for the first time ...

Wow! Right?!

And thank you for buying my e-book.
I'd love to hear about your success
stories! Or did you have trouble un-
derstanding part of my e-book? Have
some *constructive* feedback? I want to
hear from you; please share with me at
ammonymous@gmail.com.

BONUS

Here's a little extra information that
may come in handy

Resetting the Combination

You don't need a safe technician or
locksmith to reset your safe combina-
tion for you. If you're mechanically in-
clined, you may not even need a lock
change key.

If you do have a key (the right key for your lock), this is how you change your combination:

- 1) Open your safe.
- 2) With your safe door open, lock it and dial your combination *on the change index mark*, not on the open index mark.
- 3) Insert the correct combination change key in the back of your lock. Using the wrong key can foul up your lock.
- 4) Turn the dial four full turns and begin dialing your new combination
- 5) Remove the key.
- 6) Test your combination (on the open index) several times. Also, see Tip #2, above.

Alternately, you may physically dismantle your lock and reassemble it with a new combination. If your lock has a change key hole (back plate of the lock), the wheels of your lock will consist of inner and outer rings. By changing the orientation of these inner and outer rings you can change the lock combination.

ONLY do this if you are very mechanically inclined. Also, see Tip #2.

BONUS: Cracking Home Fire Safes with Electronic Locks

Walmart, Office Depot, Lowes, and other big box stores often sell home fire safes with electronic locks. Obviously, fire safes are intended to guard against fires not burglars, making many of these safes extremely easy to crack. The better you know the safe, the easier cracking it will be. Most electronic locks will fail for two reasons: the battery is dead or the battery is dead. (Sometimes the internal working get bound up a bit, so you just need to wiggle the door handle a little.) Because of the constant threat of a dead battery, electronic safes always have a backup access method. Sometimes the batteries are included on the exterior of the safe, but more commonly the safe can be unlocked with a tube key. Tube lock picks are easy to purchase, although some safes use extra deep tube locks that normal tube lock picks won't work on.

But don't worry, it's easier than that.

The basic principle of electronic safe locks is the same as mechanical dial locks: the lock merely stops the handle from turning the door lock bolts. In the case of a mechanical dial lock, a lock bolt blocks the door lever's operation. In the case of an electronic lock, the correct combination entered into the keypad powers a solenoid. The solenoid uses electromagnetic forces to draw a locking pin back, thus allowing the door handle to operate the door locking bolts. Locking pins may use gravity to stay in the locked position, or the more likely scenario is that a light spring holds the pin in the locked position. In either case you can use gravity to defeat the locking pin.

In the unlikely case a spring is not used on the locking pin, gravity is used to hold the spring down in the locked position. The solenoid pulls the pin up to open the safe. Just turn the safe upside down and it will open.

In the much more likely case that a retaining spring is used on the lock-

ing pin, the spring will probably hold the pin *up* in the locked position. The purpose of this is so the solenoid will be gravity assisted in pulling the pin down and out of the way of the door bolt mechanisms. Using gravity to assist the solenoid in drawing the pin down against its retaining spring saves battery power and increases reliability. However, this also makes defeating that electronic lock easier. Here's how:

Place the safe on a hard surface (not carpet). Tilt the safe back so the front of the safe is 3-6 inches off the floor. Let the safe fall to the floor. In the instant the safe hits the floor (or the millisecond after), the locking pin will continue its downward motion against the retaining spring, thereby unlocking the safe for a split second before the retaining spring pops the pin back into the locked position.

If you can time it during that millisecond after the safe drops to the floor, turn the door handle and the safe will open.

Once open, it's easy to replace the batter or change the electronic combination. Simply search the instructions for your model online.